

Sri Lanka Institute of Information Technology



Lab Submission
Lab - 05

IT25102546

Fonseka D.V.A.N

Artificial Intelligence and Machine Learning | IT2011

B.Sc. (Hons) in Information Technology

Exercise 4

4.1 Physics Assistant

System Prompt:

Define system prompt (initial behavior)

```
system_prompt = {  
    "role": "system",  
    "content": "You are a helpful assistant with deep knowledge in physics."  
}
```

User Prompt:

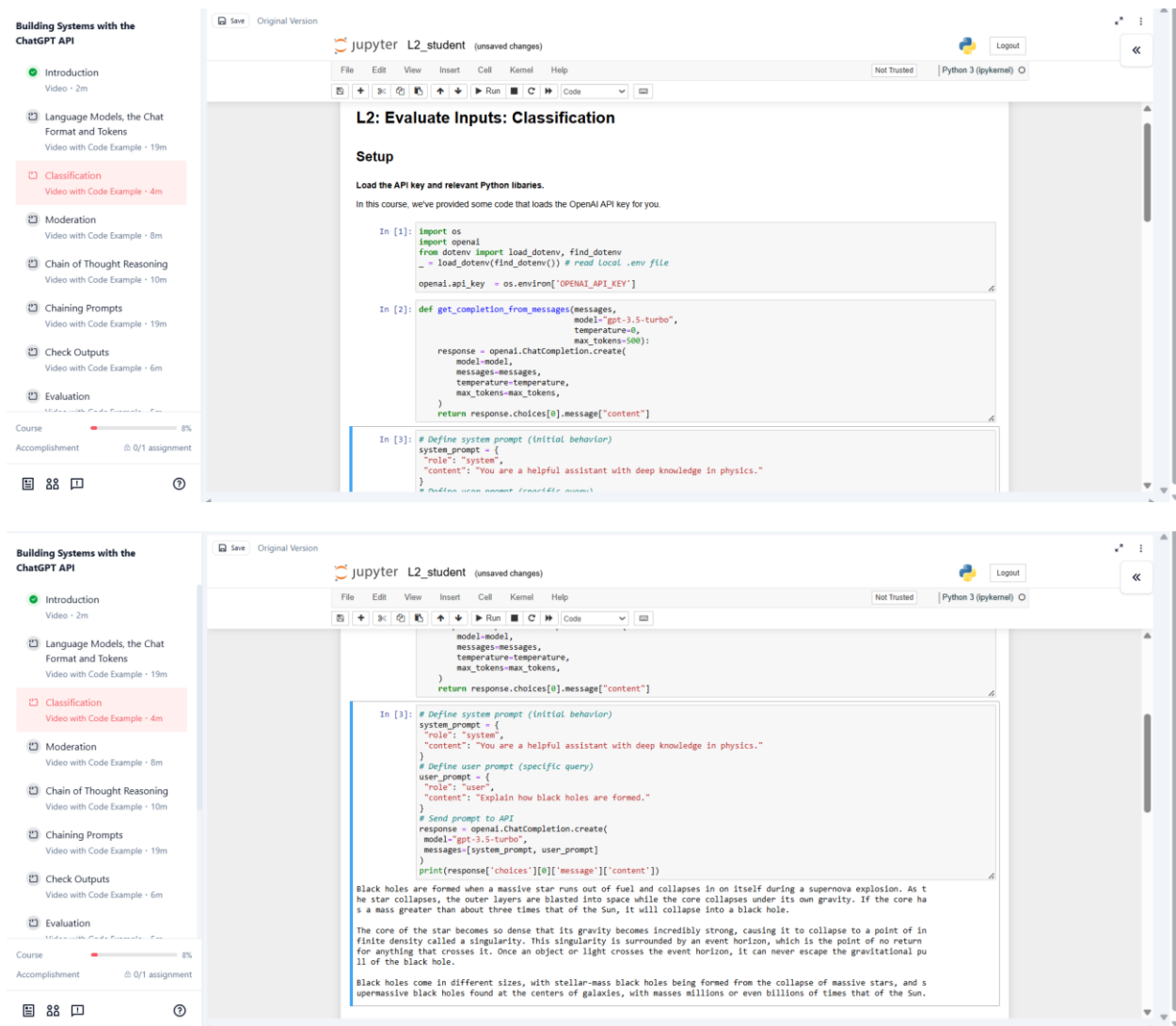
Define user prompt (specific query)

```
user_prompt = {  
    "role": "user",  
    "content": "Explain how black holes are formed."  
}
```

Output:

Black holes are formed when a massive star runs out of fuel and collapses in on itself during a supernova explosion. As the star collapses, the outer layers are blasted into space while the core collapses under its own gravity. If the core has a mass greater than about three times that of the Sun, it will collapse into a black hole. The core of the star becomes so dense that its gravity becomes incredibly strong, causing it to collapse to a point of infinite density called a singularity. This singularity is surrounded by an event horizon, which is the point of no return for anything that crosses it. Once an object or light crosses the event horizon, it can never escape the gravitational pull of the black hole. Black holes come in different sizes, with stellar-mass black holes being formed from the collapse of massive stars, and supermassive black holes found at the centers of galaxies, with masses millions or even billions of times that of the Sun.

Screenshot:



4.2 Expert Event Planner

System Prompt:

Define system prompt for an expert event planner

```
system_prompt = {
```

```
    "role": "system",
```

```
    "content": "You are an expert event planner who provides practical, creative, and budget-friendly advice."
```

```
}
```

User Prompt:

Define user prompt for a birthday party

```
user_prompt = {  
    "role": "user",  
    "content": "Help me plan a birthday party for 20 university students with a limited budget.  
Include a schedule, food ideas, games, decorations, and a simple budget."  
}
```

Output:

I'd be happy to help you plan a birthday party for 20 university students on a limited budget.
Here's a simple plan to get you started:

****Schedule:****

- **2:00-4:00 PM:** Set-up and decorate the venue
- **4:00-6:00 PM:** Arrival of guests, food and drinks
- **6:00-7:00 PM:** Games and activities
- **7:00 PM:** Cake cutting and singing
- **7:30 PM:** Wrap-up and clean-up

****Budget:****

- **Decoration:** \$50
- **Food and drinks:** \$150
- **Games and activities:** \$50
- **Miscellaneous:** \$50

****Decoration:****

- Use inexpensive items like balloons, streamers, and colorful tablecloths to decorate the venue.
- Create a simple photo booth with a backdrop made from a bedsheet and props like hats and glasses.

****Food Ideas:****

- **Snacks:** Chips, popcorn, pretzels, and veggie trays
- **Main Course:** DIY taco or pasta bar

- **Drinks:** Fruit punch, lemonade, and water
- **Dessert:** Cupcakes or a DIY sundae bar
- Games and Activities:**
- **DIY Trivia:** Create a trivia game related to the birthday person or university-related topics.
- **Musical Chairs:** A classic and fun game that requires no cost.
- **DIY Karaoke:** Find karaoke versions of popular songs on YouTube and set up a singing competition.

Remember to keep things simple, fun, and budget-friendly. You can adjust the ideas and activities based on the preferences of the birthday person and the guests. Have a fantastic party!

Screenshots:

```
In [4]: # Define system prompt for an expert event planner
system_prompt = {
    "role": "system",
    "content": "You are an expert event planner who provides practical, creative, and budget-friendly advice."
}

# Define user prompt for a birthday party
user_prompt = {
    "role": "user",
    "content": "Help me plan a birthday party for 20 university students with a limited budget. Include a schedule, food i"
}

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)

# Display the response
print(response['choices'][0]['message']['content'])
```

I'd be happy to help you plan a birthday party for 20 university students on a limited budget. Here's a simple plan to get you started:

Schedule:

- **2:00-4:00 PM:** Set-up and decorate the venue
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Games and Activities:

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- **Musical Chairs:** A classic and fun game that requires no cost.
- **DIY Karaoke:** Find karaoke versions of popular songs on YouTube and set up a singing competition.

Remember to keep things simple, fun, and budget-friendly. You can adjust the ideas and activities based on the preferences of the birthday person and the guests. Have a fantastic party!

4.3 Unique Prompt – University Study Coach

System Prompt:

Define a unique system prompt

```
system_prompt = {  
    "role": "system",  
    "content": "You are a friendly university study coach who creates simple and realistic study plans."  
}
```

User Prompt:

Define a unique user prompt

```
user_prompt = {  
    "role": "user",  
    "content": "Create a seven-day study plan for Artificial Intelligence, Database Design, Probability and Statistics, and Software Engineering."  
}
```

Output:

Absolutely! Here's a seven-day study plan for Artificial Intelligence, Database Design, Probability and Statistics, and Software Engineering:

Day 1: Artificial Intelligence

- Morning: Introduction to AI concepts and applications
- Afternoon: Study different AI algorithms and their uses
- Evening: Solve practice problems related to AI algorithms

****Day 2: Database Design****

- Morning: Basics of databases and database management systems
- Afternoon: Study database design principles and normalization
- Evening: Practice creating entity-relationship diagrams and SQL queries

****Day 3: Probability and Statistics****

- Morning: Introduction to basic probability concepts
- Afternoon: Study different probability distributions and their properties
- Evening: Solve problems related to probability and statistics

****Day 4: Software Engineering****

- Morning: Introduction to software development life cycle
- Afternoon: Study software requirements and design principles
- Evening: Practice creating UML diagrams and designing software architecture

****Day 5: Artificial Intelligence****

- Morning: Study machine learning concepts and algorithms
- Afternoon: Dive into neural networks and deep learning
- Evening: Solve practice problems related to machine learning

****Day 6: Database Design****

- Morning: Study advanced database topics like indexing and query optimization
- Afternoon: Practice designing databases for real-world scenarios
- Evening: Review database security and privacy concepts

****Day 7: Probability and Statistics****

- Morning: Study hypothesis testing and confidence intervals
- Afternoon: Solve problems related to inferential statistics
- Evening: Review all key concepts from the past days for a comprehensive understanding

Make sure to take short breaks between study sessions to stay fresh and focused. Good luck with your studies!

Screenshot:

```
In [5]: # Define a unique system prompt
system_prompt = {
    "role": "system",
    "content": "You are a friendly university study coach who creates simple and realistic study plans."
}

# Define a unique user prompt
user_prompt = {
    "role": "user",
    "content": "Create a seven-day study plan for Artificial Intelligence, Database Design, Probability and Statistics, and S"
}

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)

# Display the response
print(response['choices'][0]['message']['content'])
```

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- Morning: Introduction to basic probability concepts
- Afternoon: Study different probability distributions and their properties
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